



DBA4811 Analytical Tools for Consulting

Project Report

Group 1

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1. Introduction

1.1 Client Background

ConnectX Communications is a mid-sized telecommunications provider operating in the United States, offering a wide range of telecommunication services, including broadband (DSL and fiber-optic), bundled phone and TV packages, and mobile subscriptions for consumers. The company primarily serves both urban and suburban areas, with a strategic focus on providing high-quality connectivity and reliable customer support. Over recent years, ConnectX has observed an increase in customer churn, posing a direct challenge to its subscription revenue model.

1.2 Market and Competitor Landscape

The U.S. telecommunications market is valued at \$522.8 billion in 2025 and is projected to reach \$725.7 billion by 2030, growing at a compound annual growth rate (CAGR) of 6.9% (Grand View Horizon, 2025). Key competitors include national giants such as AT&T, Verizon, and T-Mobile. Technological upgrades like 5G and fiber expansion require heavy capital, limiting spending on retention and marketing, while customer expectations for high-speed, reliable and bundled services continue to rise.

1.3 Problem Statement

ConnectX is currently experiencing rising customer churn, which poses financial and operational challenges. Each churned subscriber reduces monthly recurring revenue, directly impacting cash flow, while the premature loss of long-term customers diminishes Customer Lifetime Value (CLV) and hence erodes overall profitability. Additionally, replacing churned customers increases marketing and acquisition costs, straining budgets and reducing the efficiency of growth initiatives.

Currently, ConnectX lacks a systematic, data-driven approach to identify high-risk customers, understand the drivers of churn and implement targeted retention strategies.

1.4 Objectives

This engagement is designed to help ConnectX proactively manage customer churn through data-driven insights. The primary objectives are as follows:

1. Identify high-risk customers: Using cluster analysis, determine the demographic, behavioral and service-related characteristics that are most strongly associated with churn.
2. Develop predictive models: Use historical customer data to forecast which subscribers are most likely to churn and highlight the key factors driving churn risk, to enable the company to intervene proactively.
3. Provide actionable recommendations: Design targeted retention strategies based on analytical insights, supporting improved revenue stability, higher customer loyalty and more effective allocation of retention resources.

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2. Data Exploration and Preparation

Our analysis covers 4,073 ConnectX customers, focusing on how demographics, billing habits, and service usage relate to customer retention and churn.

2.1 Data Preparation

We began with an exploratory data analysis. We identified a small number of new customers (11) whose 'TotalCharges' were missing because they had just joined and made no payments yet. To maintain consistency, we filled these with zero to include them in our analysis. We also standardized categorical and numerical fields to ensure reliable comparisons across customers. Once the data was cleaned, we conducted cohort, survival, and clustering analyses to understand customer profiles and retention trends.

2.2 Exploratory Data Analysis (EDA)

EDA was conducted to uncover key drivers of churn, building a high-level view of *who* is leaving, *why* are they leaving, and *when* churn is most likely to occur. (See Appendix 1 for all graphs.)

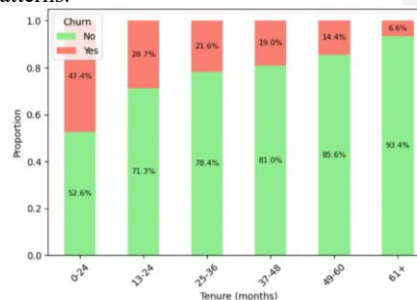
There are several customer attributes that contextualise churn behaviour. When segmenting by **payment method**, customers paying by electronic checks show the **highest churn rate at ~45%**, suggesting that **payment experience influences retention**, as friction in the payment process causes dissatisfaction. It also reflects **demographic differences**. Customers with lower digital literacy or trust in online payments are more likely to disengage when facing billing complexity. Churn rates vary by **internet service type**. Fibre-optic customers, a higher-value segment, record a **high churn rate of 42%** (Figure 2.2), suggesting higher expectations or lower tolerance for perceived value gaps. Conversely, customers without internet services show **lowest churn rate at just 7%**, indicating stronger stability among low-engagement users. **Senior customers show a lower churn rate (20%) compared with younger users (42%)**, indicating stronger loyalty and higher inertia once a stable relationship is formed. Younger customers, in contrast, appear more likely to leave. These patterns highlight that churn is also influenced by demographic characteristics. Taken together, churn is driven by both customer characteristics and structural factors.

2.2.1 Cohort Analysis

We now turn to cohort analysis to examine *when* churn occurs and how ConnectX's tenure, contract types, and pricing segments influence retention patterns.

2.2.1.1 Tenure Segmentation

Segmenting customers by tenure answers key questions about *when* customers are most likely to churn and how tenure affects contract and retention dynamics. There is a strong tenure effect on churn behaviour, with churn concentrated in the **early lifecycle**. Nearly half (47.4%) of customers leave **within their first year of**



subscription (Figure 2.4), indicating that the onboarding and early engagement phase is key to retention. Beyond This stage, churn rates decline sharply: customers who remain for more than five years exhibit a 6.6% churn rate.

2.2.1.2 Contract Type Segmentation

Tenure is linked to contract type. Month-to-month (MTM) customers have a median tenure of 10 months; one-year and two-year contract holders stay for around 45 and 65 months, respectively (Figure 2.5). This pattern emphasises how contract structure strongly influenced tenure length, likely because flexible contracts attract short-term, price-sensitive users; longer-term

Figure 2.4 Churn Rate by Tenure

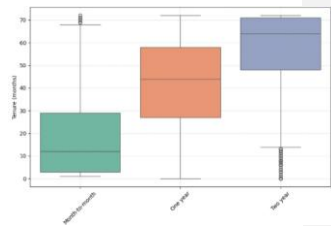
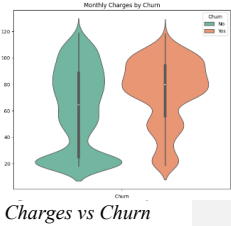


Figure 2.5 Boxplot of Tenure across Contract Types

2.2.1.3 Price Segmentation

Figure 2.6 shows that churned customers tend to pay around \$60–\$65, while retained customers cluster in two price ranges: \$20–\$30 and \$75–\$95. This suggests that customers generally perceive these ranges as a “fair price,” with payments within these bands associated with higher retention. Pricing significantly influences retention, especially for newer customers.

As shown in Figure 2.7, customers in their first two years paying mid-to-



Charges vs Churn



Figure 2.7 Heatmap of Churn Rate vs Tenure, Monthly Charges

Contract type reinforces this pattern (Figure 2.8): MTM customers pay the highest median charge (\$73.2) but have the shortest median tenure (10 months). One- and two-year contract holders, pay \$64.3 to \$68.8 and remain for 45 to 65 months. This demonstrates that discounted long-term contracts convert price concessions into loyalty, while flexible contracts attract short-term, price-sensitive users.

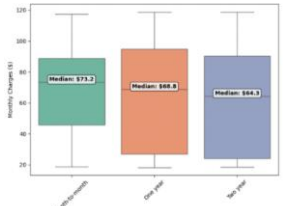


Figure 2.8 Boxplot of Monthly Charges across Contract Types

2.2.1.4 Internet Service Segmentation

Building on the earlier analysis, we examined churn patterns by Internet Service, revealing a critical early period of roughly 20–30 months where churn is concentrated across all users. Notably, DSL non-churners also cluster within this window, suggesting that customers who navigate the initial high-churn phase are more likely to remain engaged. This highlights the importance of

targeted retention strategies during the early tenure months. The overall pattern is consistent across all internet service types, with churn predominantly occurring within the first two years. Overall, these findings indicate that price tolerance grows with tenure, while early-stage, high-paying customers are the most vulnerable, highlighting the first year as the critical window for engagement and ensuring perceived value matches pricing.

2.4 Customer Retention Analysis

2.4.1 Survival Analysis Methodology

Survival analysis is a statistical approach originally used in medicine and engineering to estimate how long subjects “survive” before an event occurs. In our context, the “event” is customer churn; “survival time” represents how long a subscriber stays before churning. Using the Kaplan-Meier method, the analysis measures the probability that a customer remains active over time, highlighting when churn risk peaks and how retention varies across different customer segments.

2.4.2 Retention Rate Patterns

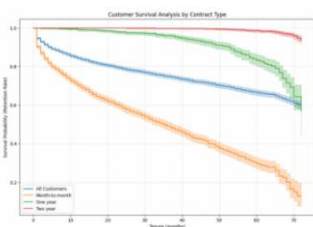


Figure 2.8 Kaplan-Meier Survival Curve by Contract Type

Figure 2.8 illustrates that retention declines steeply within year, the first 6 to 12 months, reaffirming the first year as the critical period for customer engagement. Month-to-month over (MTM) customers exhibit the fastest churn, with only ~70% five surviving the first year. One-year contract holders are more stable initially (~97% survival during the contract) but experience a sharp “cliff” when the contract expires, dropping to ~90% and gradually declining to ~75% by month 60. Two-contracts provide the strongest retention survival remaining above 95%

years, highlighting the protective effect of longer-term commitments and the contrast in churn behavior across contract types.

Figure 2.9 examines MTM customers across pricing tiers, revealing that low-spending customers (blue) exhibit higher retention than med- (orange) and high-spending (green) peers. This suggests that early-stage, higher-value customers are particularly sensitive to perceived value, making them more likely to churn if the service does not meet expectations.

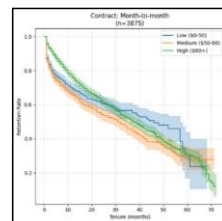


Figure 2.9: Kaplan-Meier Survival Curve for MTM Customers by Price Tier

These findings show that tenure and pricing jointly shape churn. Early-stage, medium- and high-paying customers, especially on MTM plans, are most at risk, reflecting perceived gaps between price and value. MTM subscribers also pay more on average, a structural effect of flexible contracts attracting short-term, less loyal customers. By

contrast, longer-term contracts, though often lower in monthly charges, anchor loyalty and reduce churn, highlighting that contract design and pricing together drive retention outcomes.

2.5 Customer Clustering Analysis

K-means clustering was applied to segment ConnectX customers based on behavioral and financial attributes. This unsupervised approach allows the data to reveal natural groupings and patterns in churn behaviour, producing distinct clusters with varied churn probabilities and risk profiles.

2.5.1 Clustering Methodology

We iteratively tested different numbers of clusters (see Appendix) to identify the configuration that produced the clearest and most actionable distinctions. A four-cluster solution was selected, as it highlighted markedly different churn rates while capturing meaningful variations in customer attributes and risk profiles.

2.5.2 Risk Segment Profiles

Figure 2.10 and 2.11 illustrate how the clusters differ in customer tenure and monthly spending patterns.

Cluster 0 (Moderate Risk): This cluster makes up 9.7% of customers, of which 24.9% churned. This group has an average tenure of ~32 months, pays moderate monthly charges (~USD42), and has moderate adoption of online services (~40% use OnlineSecurity, OnlineBackup, or TechSupport). Slightly more than half hold month-to-month contracts (55%), and 15% are senior citizens.

Cluster 1 (Low Risk): 29.8% of customers are in this cluster, of which only 15.9% churned. This cluster represents long-tenure (~57 months), high-value customers with high monthly charges (~USD92), high service adoption (~67%), and predominantly one- or two-year contracts; 20% are senior citizens. These customers are loyal and stable, showing the lowest churn among high-value segments.

Cluster 2 (Very Low Risk): Comprising 21.7% of customers, of which 7.4% churned, this cluster constitutes low-value, low-risk customers with short tenure (~31 months), very low monthly charges (~USD21), minimal service usage, and mostly two-year contracts (~42%). They are largely non-senior and exhibit the lowest churn overall.

Cluster 3 (High Risk): This is the largest (38.8% of customers) and most volatile segment (45.8% churn rate). This cluster is characterized by very short tenure (~14 months), high monthly charges (~USD74), low service adoption (~25–34%), predominantly month-to-month contracts (88%), and

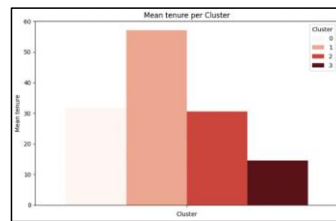


Figure 2.10 Mean Tenure per Cluster

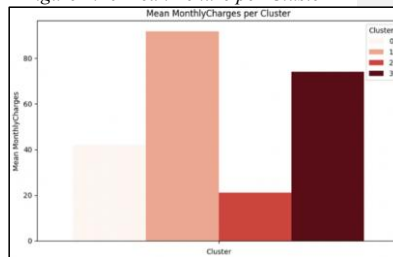


Figure 2.11 Mean Monthly Charges per Cluster

a slightly higher proportion of senior citizens (~21%). This group faces the highest churn risk, making them a priority for early retention efforts.

A summary of the service engagement of each cluster is listed below:

Service (Churn rate)	Cluster 0 (24.9%)	Cluster 1 (15.9%)	Cluster 2 (7.4%)	Cluster 3 (45.8%)
Internet	DSL only	DSL (~33%) Fiber optic (~67%)	No Internet	DSL (~38%) Fiber optic (~62%)
Phone	No phone service	Yes	Yes	Yes
(Median) No. of services engaged	5	6	1	3
Contract Type	~55% monthly ~21% 1-year ~24% 2-year	~27% monthly 33% 1-year ~40% 2-year	~34% monthly ~24% 1-year ~42% 2-year	~88% monthly ~10% 1-year ~2% 2-year

To simultaneously assess both churn risk and financial contribution, a Cluster Risk-Value Matrix (Figure 2.12), is plotted, clustering by churn rate and customer value, with value quantified as churn-adjusted total charges ($Total\ Charges \times (1 - Mean\ Churn\ Rate)$). This adjustment accounts for probability of churn, providing a realistic estimate of expected revenue of a cluster. The visualization reinforces earlier patterns: Cluster 1 occupies high-value, low-risk quadrant, Cluster 3 occupies top-right quadrant, combining high value with high churn risk. We hypothesise that targeting retention efforts at Cluster 3 is therefore likely to deliver the greatest financial returns for ConnectX.

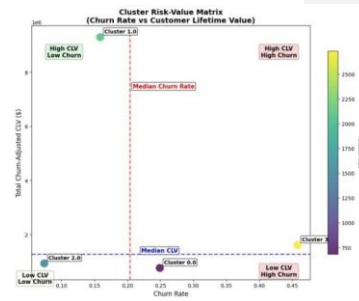


Figure 2.12 Cluster Risk-Value Matrix

3. Predictive Modelling

To combat churn and reduce revenue loss, we developed a predictive model that quantifies individual customer risk and key drivers, enabling ConnectX to proactively direct retention strategies toward the most vulnerable segments.

3.1 Additional Data Preparation

Firstly, we build upon the cleaned and explored dataset, applying further processing steps to optimize model performance and interpretability. To summarize the process for brevity, we addressed skewness via square root transformations and eliminated perfect collinearity by consolidating service categories. Business-meaningful correlations were preserved to enhance predictive signals while maintaining model integrity (Appendix 2).

3.2 Methodology

3.2.1 Evaluation Strategy

In predicting customer churn, missing a customer who is likely to leave (False Negative, FN) can result in lost revenue and reduced CLV. Therefore, we **prioritized recall**

$\left(\frac{\text{correctly predicted churners (TP)}}{\text{all actual churners (TP + FN)}}\right)$ as our primary evaluation metric to ensure the model identifies as many at-risk customers as possible. To balance the risk of misidentifying non-churners as churners, we also considered AUC-ROC as a secondary metric, which evaluates how well the model distinguishes churners from non-churners, particularly crucial as only 26% of the customer records were churners. This helps the company target retention efforts while minimizing wasted resources.

3.2.2 Feature Engineering

Evaluating Impact of Transformed Numerical Feature: To assess whether transforming the TotalCharges column truly benefitted model performance, we applied cross-validation on the training set using a baseline logistic regression model. The transformed feature yielded a mean recall of 0.53, lower than 0.55 for the original column. Given the lack of improvement and the poorer interpretability, we retained the original TotalCharges feature for subsequent steps.

Adding New Features: Based on the insights gleaned from Section 2, we created the following features (Figure 3.1) to capture potential interactions and factors that may impact a customer's likelihood to churn.

has_family	• Customers who have family members	monthly_charge_per_svc	• Average monthly charge per service
young_monthly	• Customers on monthly contracts who are young	has_streaming	• Customers who get streaming services
non_monthly	• Non-monthly contract type	auto_payment	• Customers who make automatic payment transfers
num_services	• Total no. of services a customer receives	internet_noaddons	• Customers who use Internet services and do not have add on services
phone_and_internet	• Customers who have both phone and internet service	pct_of_contract_completed	• Percentage of contract completed

Figure 3.1 New features added

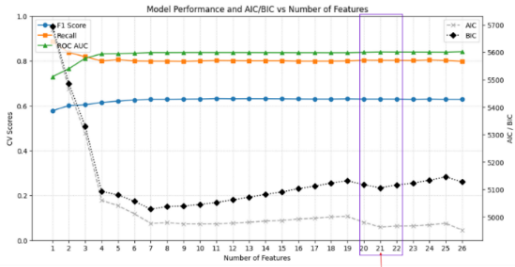
Feature Selection: We used two complementary approaches to guide feature selection for the churn model: forward selection and Lasso (L1) regularization.

In forward selection, features are added one by one, prioritizing those that improve model performance while keeping the model as simple as possible. We also used AIC and BIC as supplementary metrics as they collectively measure predictive fit against model complexity (Figure 3.2). The second method, Lasso regularization shrinks less informative feature coefficients toward zero to highlight the features most relevant for predicting churn. We iteratively testing different regularization strengths to select the features that optimized both recall and ROC-AUC (Figure 3.3).

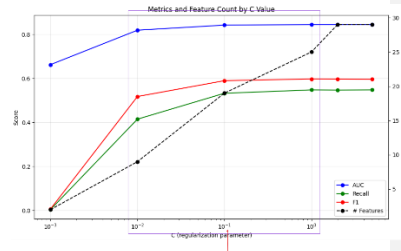
Comparing the two approaches, we selected the feature set identified by forward selection, as it delivered better recall (0.803 vs 0.547), lower variance, and a more compact set of features (21 vs 25), which would result in a simpler model.

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Optimal no. of predictors: 21
Mean CV Recall: 0.80 ± 0.02 ; Mean CV AUC: 0.84
 Figure 3.2 Stepwise performance of forward feature selection



Optimal no. of predictors: 25
Mean CV Recall: 0.55 ± 0.03 ; Mean CV AUC: 0.84
 Figure 3.3 Performance of Lasso Regularisation

3.2.3 Resampling

Using the training set restricted to the selected features, we applied **resampling techniques to address class imbalance**, where non-churning customers outnumber churning ones. Resampling helps the model **better learn patterns associated with the minority class (churners)**, improving predictive performance. We tested two approaches. The first method, SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic examples of the minority class to balance the dataset, whereas the second method, SMOTE-ENN (SMOTE-Edited Nearest Neighbours) combines SMOTE with ENN, which additionally removes ambiguous or noisy examples.

	Mean Recall	Standard Deviation
No resampling	0.803	0.0172
SMOTE	0.796	0.0145
SMOTE-ENN	0.813	0.0157

Figure 3.4 Rebalanced Train Set Performance

Comparing the original imbalanced training set with the resampled, **SMOTE-ENN achieved highest mean recall (0.813)** (Figure 3.4).

Given its performance, we chose

SMOTE-ENN.

3.2.4 Model Fitting and Tuning

To identify the best classifier, we evaluated **10 models using cross-validation and hyperparameter tuning** (See Appendix 3). These included **linear and nonlinear approaches** such as bagging and boosting ensembles, distance-based methods, support vector machines, and neural networks (Appendix 4). Trying multiple methods ensures that the observed predictive performance is robust. We present the three top-performing models here.

Method	Model	Model Performance		Non-Churn Misclassification
		Test	Train	
Linear	Tuned L1 Logistic Regression	Recall: 0.88 AUC: 0.84	Recall: 0.88 AUC: 0.84	
Boosting	Tuned AdaBoost	Recall: 0.88 AUC: 0.79	Recall: 0.89 AUC: 0.79	X
Boosting	Tuned XGBoost	Recall: 0.84 AUC: 0.83	Recall: 0.86 AUC: 0.83	

The tuned L1 **Logistic Regression model** with features selected via forward selection was selected as the

best model. Although AdaBoost attained comparable recall, logistic regression demonstrated superior AUC (0.84 versus 0.79), indicating a **lower likelihood of incorrectly classifying non-**

churning customers, while also providing **enhanced interpretability** for business stakeholders. Furthermore, the model exhibits minimal overfitting, making it **reliable for predicting churn of new customers**.

3.2.5 Model Interpretation

The L1-regularized logistic regression model retained 13 key predictors that significantly influence churn. Using their coefficients (Appendix 5) and the derived odds ratios ($e^{coefficient}$), we quantify how much each factor affects churn risk relative to customers without the attribute.

More Likely to Churn		Less Likely to Churn		
Attribute	Odds Ratio	Attribute	Odds Ratio	% Less likely to churn ¹
Tenure	4.19	No Internet service	0.48	52
Month-to-month contract	3.03	Receives tech support	0.75	25
Fibre optic Internet	2.74	Has online backup	0.83	17
Pays via electronic check	1.61	Has phone service	0.87	13
Senior citizen	1.29			
Internet with no add-ons	1.18			
Higher average monthly charge per service	1.18			

Figure 3.5 Odds Ratios of Key Characteristics Affecting Churn Likelihood

For example, customers on month-to-month contracts are 3× as likely to churn as those on longer-term contracts. Conversely, attributes on the right can be interpreted using the **reduction in churn risk**. For instance, customers without internet service are 52% less likely to churn than those with. It is important to note **not all predictors are actionable levers**. Some are interconnected, so **the effect of a single factor alone may not directly translate into a change in churn**. Only certain variables represent true levers that can be influenced to reduce churn. Others primarily reflect underlying customer behavior and patterns, which we will elucidate.

Supplementing results from the best linear model, we examine the nonlinear XGBoost model that performed well on both metrics and slightly outperformed AdaBoost in AUC. This captures robust signals across modeling techniques to ensure insights are not artifacts of a single method.

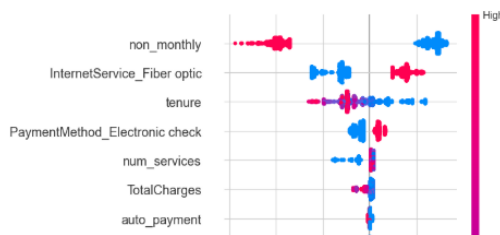


Figure 3.6 Cropped SHAP Analysis
(For full analysis, see Appendix 6)

We performed a SHAP analysis on the tuned XGBoost model (Figure 3.6) to interpret its predictions. The plot highlights the most influential features driving churn risk, ranked by overall impact. For each feature, clusters on the right (positive SHAP values) indicate higher predicted churn risk, and vice versa. Both XGBoost and Lasso converge on key

predictors such as contract type, tenure, internet service type, and payment method. XGBoost

¹ % less likely to churn = $(1 - \text{Odds Ratio}) \times 100\%$

additionally shows customers with more subscribed services and automatic payments are less likely to churn. By considering both the L1-regularized logistic regression and XGBoost models, we identified key insights.

Month-to-Month Contract Type and Early Churn: Each additional month of tenure reduces a customer's likelihood of churn, highlighting the retention benefits of longer relationships. Customers on month-to-month contracts, however, are 3.0× as likely to churn compared to those on one- or two-year contracts. As month-to-month customers naturally have shorter tenure, tenure and contract type are closely linked. Retention efforts should thus prioritize early engagement and contract incentives, leveraging actionable levers to reduce churn among new or short-tenure customers.

Greater Service Breadth and Engagement: Both models indicate that customers using more services have lower churn risk, with each additional service reducing churn odds by roughly 20%. This effect is closely linked to average monthly charges per service: higher engagement not only increases perceived value but also anchors customers through cross-selling and bundling. Customers with minimal service adoption or paying higher charges per isolated service remain more vulnerable, particularly in early tenure, underscoring the need to align perceived value with pricing. This confirms that service adoption is a strong predictor of retention, suggesting that in many regions, fiber optic performance does not meaningfully exceed DSL for typical household

Potential Fiber Optic Service Dissatisfaction: Subscribers with fiber optic internet are 2.7× as likely to churn as those using DSL or with no internet subscription, suggesting potential dissatisfaction with pricing or service reliability. Further analysis shows that among churners, the highest proportion of seniors occurs within the fiber optic segment, highlighting a high-value but at-risk subgroup, corresponding with the high-risk Cluster 3 identified earlier. Moreover, analysis of network infrastructure across U.S. markets

suggests that in many regions, fiber optic performance does not meaningfully exceed DSL capabilities for typical household usage patterns (Dine & Kane, 2022), which may create a perceived mismatch between value and price.

Payment Method as a Marker: Customers paying by electronic check appear 1.6× as likely to churn, but this largely reflects their underlying contract type (Figure 3.8) and engagement—month-to-month or low-service customers are more likely to pay by check. In this sense, the payment method acts as a marker of higher-risk customers rather than a causal factor.

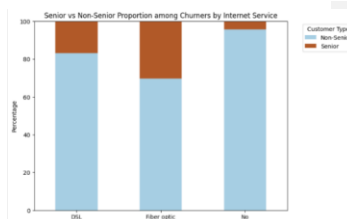


Figure 3.7 Proportion of Senior Churners by Internet Type

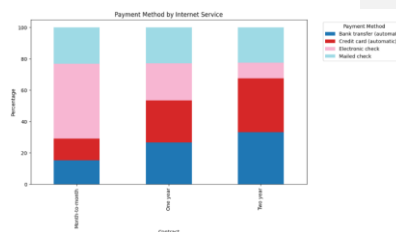


Figure 3.7 Proportion of Senior Churners by Internet Type

While it cannot be changed directly, it allows ConnectX to identify customers who may benefit from targeted retention efforts, such as early engagement, service bundling, or personalized incentives.

4. Recommendations

Based on cluster segmentation and predictive modeling, we propose a targeted retention strategy concentrating resources on segments offering highest return on investment.

4.1 Customer Cluster Profiles and Strategic Prioritization

The cluster analysis revealed four distinct customer segments with differing churn risks, lifetime values, and strategic implications. The table below summarises each segment's profile and prioritisation, based on their revenue impact, churn risk, and strategic importance to the business.

Cluster	Size	Churn Rate	ARPU*	CLV	Churn-Adjusted CLV/ customer	Revenue (Total Charges)	Risk Level	Short-term Priority	Long-term Priority
Cluster 0	9.7% (683)	24.9%	\$42	\$1495	\$1122	\$1.0M	Moderate	#2	#3
Cluster 1	29.8% (2,099)	15.9%	\$92	\$5255	\$4420	\$11.0M	Low	#3	#2
Cluster 2	21.7% (1,528)	7.4%	\$21	\$661	\$612	\$1.0M	Very Low	#4	#4
Cluster 3	38.8% (2,733)	45.8%	\$74	\$1089	\$590	\$3.0M	High	#1	#1

*Average Revenue Per User (ARPU) is derived from Monthly Charges/Total Customers in Cluster. ARPU represents the immediate monthly recurring revenue (MRR) generated by a customer in that segment.

Cluster 3's extreme churn rate, high proportion of customers and actionable intervention opportunities make it the priority for retention investment in both the short-term and long-term. Cluster 1 represents the largest total revenue base, making it a priority for long-term protection and retention effort. Cluster 0 presents moderate risk and would require short-term intervention (though to a lesser degree than Cluster 3) to mitigate churn risk factors.

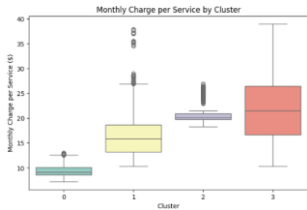
4.2 Financial Impact Methodology

We estimated each intervention's impact by converting predictive odds ratios into expected churn reduction or ARPU increase for each customer segment. We incorporated uncertainty using the 95% confidence intervals of the odds ratios to determine a realistic financial impact range.

4.3 Cluster 3: Immediate & Long-term Intervention

4.3.1 Segment Characteristics and Root Cause Analysis

Cluster 3 represents ConnectX's most critical retention challenge and simultaneously the greatest opportunity for impact, comprising nearly 40% of the customer base but contributing disproportionately to churn with a 45.8% attrition rate. These customers are



primarily new, month-to-month subscribers (88%) paying an average of \$74 monthly for fiber-optic internet with minimal adoption of additional services, resulting in the highest per-service charge. Predictive modeling confirms that month-to-month low service adoption are all strong drivers of churn, highlighting the initiatives in both short- and long-term horizons.

This segment aligns with industry profiles of “cord-cutters”: customers who already subscribe to streaming services (e.g., Netflix, Spotify) and perceive internet connectivity as a commodity rather than a differentiating factor. Adoption of value-added offerings like tech support or device protection is low not due to unawareness, but because these services do not match their needs, leaving switching costs minimal. Combined with month-to-month contracts, these factors create continuous churn vulnerability as customers make active continuation decisions monthly.

To address this, flexible, a la carte options allow customers to choose only the services they value, avoiding payment for unwanted add-ons, which increases perceived value and reduces churn risk. These insights were derived from a combination of cluster analysis, which identified demographic and usage patterns, and predictive modeling, which quantified churn risk for each attribute. Together, they guided our approach.

4.3.2 Targeted Intervention Strategy

The intervention strategy for Cluster 3 must address both the immediate churn risk and the underlying value perception challenge. We propose an approach that focuses on converting month-to-month fiber customers, with low service engagement, to 1-year contracts while offering targeted add-ons:

Intervention Tactic	Mechanism	Derived Odds Ratio	Est. Churn Reduction
Short-Term Complimentary Service Trials	Auto-enroll in tech support & backup for 60 days	0.75	15% to 20%
Contract Conversion	Lock current pricing for 1-year commitment & provide free service upgrades	3.03	9% to 20%
Flexible, A la Carte Add-On Options	Allow customers to pick the add-on services they actually want without paying for unnecessary features/ bundles	1.18	8% to 20%
30-60-90 Day Check-ins	Proactive onboarding engagement touchpoints		6% (Tridens Technology, 2025)
Senior Support	Senior Support	1.3	4% to 10%

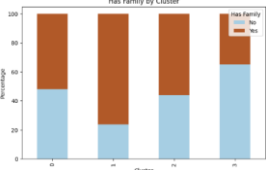
Commented [LC5]: Does this mean it's an opt-out kind of service? Or are we saying ConnectX must implement this? I think "Forced" is a funny adjective

Commented [NN6R5]: ok i change to complimentary AHAHAHAH

Using the maximum estimated impact from these set of interventions, the combined intervention strategy for Cluster 3 expects a churn reduction of **13%** (minimally 6% and up to 20%), based on the maximum estimated impact from the set of interventions.

4.4 Cluster 1: Protection and Expansion

Cluster 1 represents ConnectX's most valuable segment with \$11.0M revenue base, \$2,525 per-customer CLV and a 57-month mean tenure. Despite a relatively lower churn of 15.9%, this segment holds the highest absolute exposure, making it a strategic priority for long-term retention efforts. As these customers are already satisfied, we propose loyalty reinforcement and premium expansion rather than defensive retention, with the primary goal of preventing competitive poaching while deepening customer loyalty. The segment's profile of long-standing, high-



engagement customers largely composed of families (~80%) — suggests they are receptive to loyalty reinforcement and premium experiences rather than defensive retention measures. High service adoption indicates openness to additional offerings, particularly family-oriented or premium add-ons.

Based on industry best practices in telecom and subscription services, interventions such as loyalty pricing, VIP support, early access, and premium upsells have been shown to increase satisfaction, strengthen commitment, and reduce the risk of competitive poaching. While empirical data for these specific interventions in Cluster 1 is unavailable, industry benchmarks are used to quantify their impact.

Intervention Tactic	Mechanism	Evidence-based Est. Churn Reduction
Loyalty Pricing	5-10% discount for 5+ year customers, auto-applied	Mid-size telecom case: loyalty discounts increased retention by ~8% (Tang, 2025)
VIP Support	Priority queue, dedicated agents, <15min response SLA	Premium, experiential benefits improve engagement and loyalty (Boudet et al., 2020)
Early Access	New services/features before general availability	Enhances perception of exclusivity and innovation, driving engagement
Premium Upsells	Home automation, business-grade fiber, targeted family plans	UK telecom study: tailored add-ons can increase adoption and perceived value; supports potential ARPU lift of 5–10% (Mumba, 2021)

Given the segment's low churn and lack of historical intervention data, we focus on potential ARPU uplift as a tangible measure of impact, which captures incremental revenue per customer. Drawing industry benchmarks, we assume an ARPU increase of 5–10% from the interventions. For our projections, we use the midpoint of this range (7.5%) to estimate expected revenue gains. Thus, allowing the segment to maintain <16% churn, while providing expansion in CLV.

4.5 Cluster 0: Moderate Intervention & Steady Engagement

Cluster 0 represents moderate-risk, moderate-value, internet-only customers with average tenure (~32 months) and moderate ARPU (~\$42). Service adoption is moderate (40%), and a slight majority are month-to-month (55%), with 15% being seniors. While not a high-risk segment, the churn rate of 24.9% and \$1M revenue risk indicate measured retention efforts can yield meaningful impact.

Interventions should focus on encouraging contract upgrades, increasing digital service and phone line adoption, and reducing engagement barriers for seniors, using moderate, cost-effective tactics rather than aggressive campaigns.

Intervention Tactic	Mechanism	Derived Odds Ratio	Est. Churn Reduction
Contract Conversion	10 – 15% discount for MTM → 1-year contracts	3.03	9% to 20%
Service Trials	30-day trials (shorter than Cluster 3's 60-day)	0.75	15% to 20%
Optional Phone Line Bundling	Optional phone service add-on for interested customers	0.87	3% to 6%

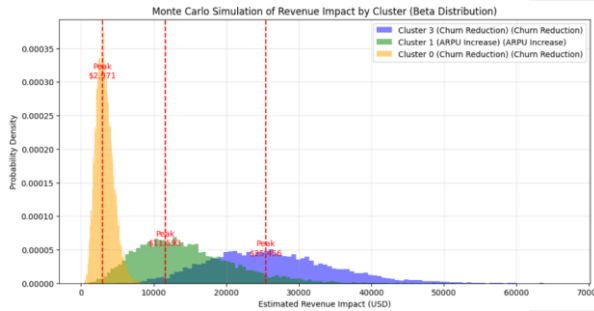
Similarly, we expect the combined intervention strategy for Cluster 0 to reduce churn by **11.5%** (minimally 3% and up to 20%), based on the estimated impact from the proposed interventions.

4.6 Financial Impact Summary

The estimated impact for each cluster was calculated with the formula *average revenue per user (ARPU) × no. of customers in the cluster × the expected effect of the intervention*. For Clusters 0 and 3, the interventions focus on churn reduction, so the estimated impact represents revenue retained from customers who would otherwise have churned. For Cluster 1, the intervention targets an increase in ARPU, so the impact represents incremental revenue gained from higher spending per customer. For each cluster, we also calculated a range of potential impact reflecting the minimum and maximum expected effectiveness of the intervention.

Cluster	No. of Customers	ARPU	Expected Improvement	Estimated Impact
3	2,733	\$74	13% (6-20% less churn)	\$26.3K (\$6-\$40K)
0	682	\$42	11.5% (3-20% less churn)	\$3.3K (\$1-5K)
1	2,102	\$92	7.5% (5-10% ARPU rise)	\$14.5K
0	682	\$42	11.5% (3-20%)	\$3.3K (\$11.6-\$39K)

To account for potential uncertainty in customer responses to the interventions, we used a Monte Carlo simulation with a Beta distribution to estimate the potential impact of targeted interventions for each customer cluster. Monte Carlo allows us to account for uncertainty by simulating 10,000 possible outcomes per cluster, producing a realistic range of revenue gains rather than a single estimate. The



Beta distribution is ideal for modeling percentages, such as churn reduction or ARPU increase. The results indicate Cluster 3 is expected to save \$26,195 (± \$8,587), Cluster 0 is expected to save \$3,302 (± \$1,233), and Cluster 1 is projected to generate \$14,468 ± \$6,776 in additional revenue. This affirms our hypothesis that interventions in Cluster 3, the high-churn, high-value segment, should be the primary focus for ConnectX, offering greatest potential revenue impact.

5. Conclusion

ConnectX grappled with critical, early-stage customer churn, primarily driven by structural issues like the high volatility of Month-to-Month (MTM) contracts and operational friction from Electronic Check. By deploying advanced analytics, including predictive modeling (L1 Logistic Regression) and a Risk-Value Cluster Triage, this created a targeted, financially driven retention strategy. We recommend ConnectX complement our predictive modeling solution with agile test-and-learn loops (Jain & Surana, 2017) to experiment with different offer designs, value propositions, and delivery modes across microsegments within clusters through A/B testing. To realize the growth projected in our playbook, embedding this agile experimentation layer will ensure that our strategies remain adaptive, evidence-driven, and resilient to real-world uncertainties that may not be fully captured by our analysis.

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Appendix 1: Exploratory Data Analysis

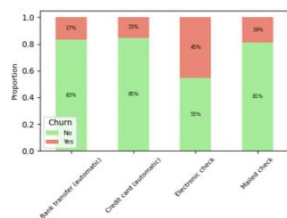


Figure 2.1 Churn Rate by Payment Method

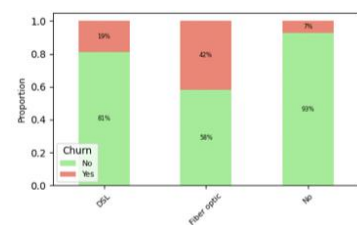


Figure 2.2 Churn Rate by Internet Service Type

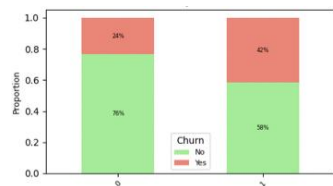
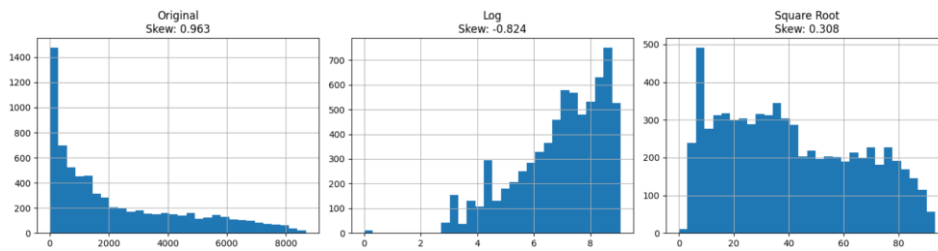


Figure 2.3 Churn Rate by Age Group (1=Senior Citizen)

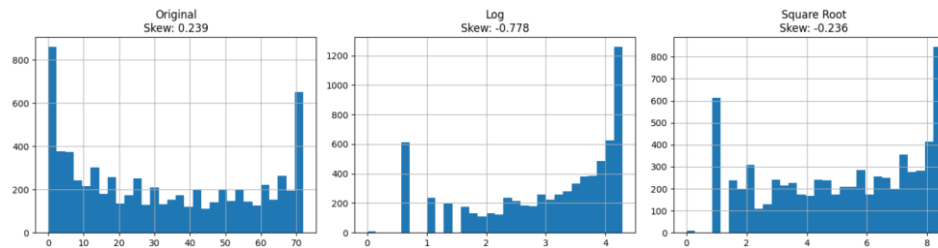
Appendix 2: Further Data Processing for Machine Learning

Normalising Distribution of Numerical Variables: The numerical variables (tenure, MonthlyCharges, and TotalCharges) exhibited skewness, which can affect the performance of models that assume approximate normality, such as logistic regression. To address this, a square root transformation was applied to the TotalCharges variable, which showed a notable improvement in **distribution symmetry**.

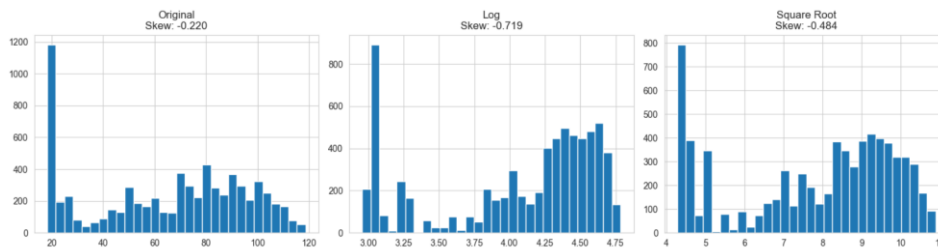
Transforming 'TotalCharges' Column:



Transforming 'Tenure' Column:



Transforming 'MonthlyCharges' Column:

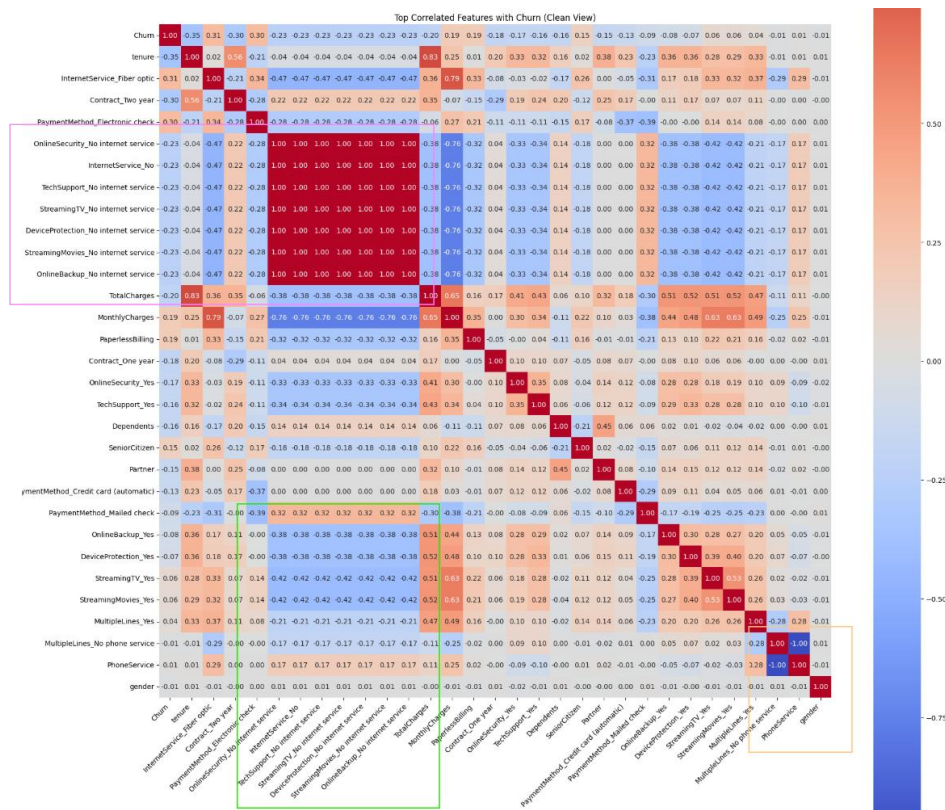


The transformation was not applied to tenure and MonthlyCharges, as the reduction in skewness was minimal and came at the expense of interpretability.

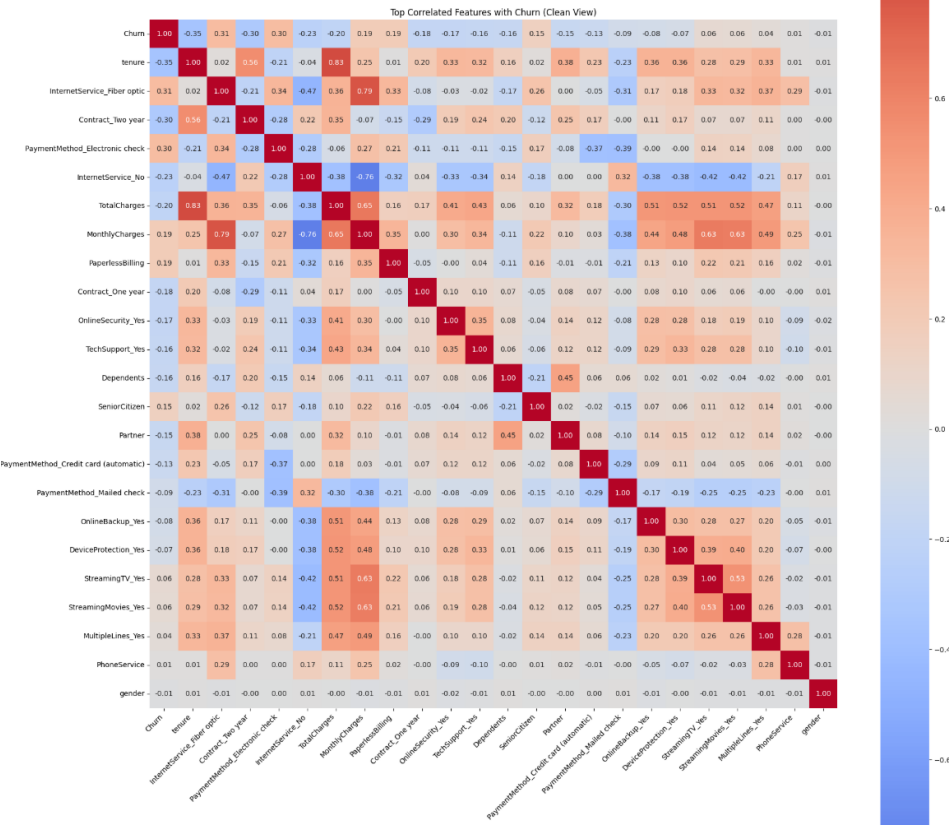
Addressing Perfect Collinearity in Service Feature Columns: The *InternetService* column and its related online service features (*OnlineSecurity*, *OnlineBackup*, *DeviceProtection*, *TechSupport*, *StreamingTV*, *StreamingMovies*) are perfectly collinear due to the 'No internet service' category. This reflects valid business logic: customers without internet service naturally do not subscribe to these add-ons. However, it introduces redundancy that can distort model estimation. A similar issue exists between *PhoneService* and *MultipleLines*. To resolve this, we **replaced** the '*No internet service*' and '*No phone service*' fields with '*No*', since the parent variables already capture this information to preserve meaning to **eliminate perfect collinearity**.

Correlation between Features: Some feature pairs show moderate to high correlation, but these reflect genuine business logic rather than redundancy. For instance, fibre-optic users pay higher fees, and longer contracts align with longer tenure. While such correlations would need to be addressed in a causal analysis to isolate the effect of a single feature, they are **valuable for predictive modeling**, as they help the model capture meaningful signals about churn risk.

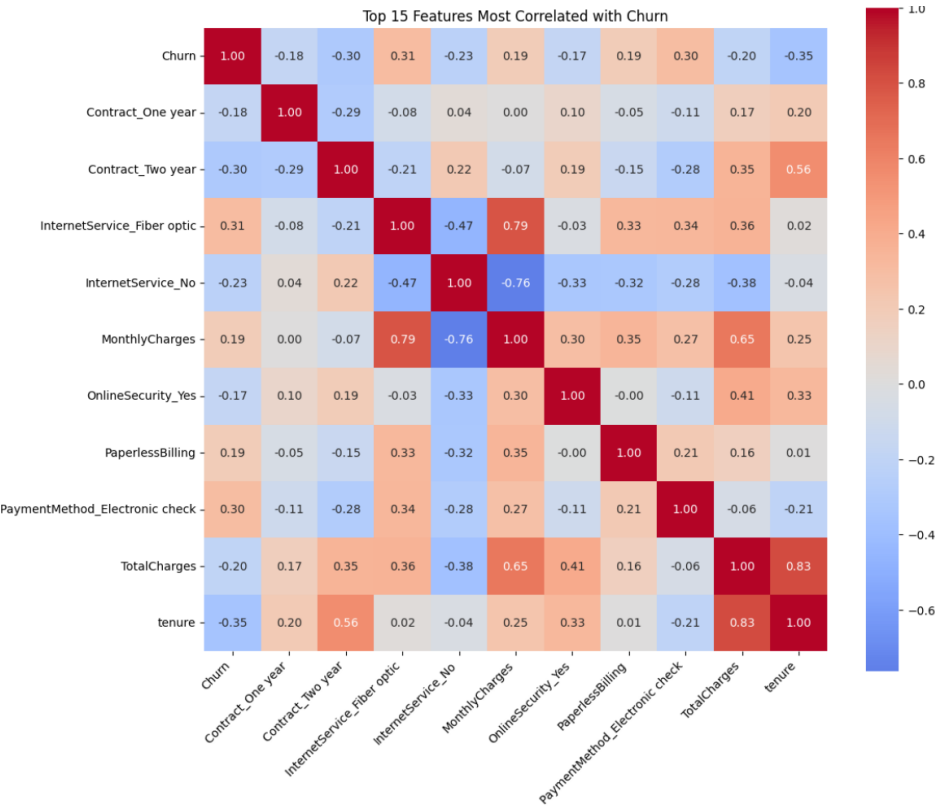
Before processing:



After processing:



Heatmap of Top 15 Features Most Correlated with Churn



High-Correlation Feature Pairs and Business Interpretation

Feature Pair	Correlation	Business Insight
InternetService_Fiber optic & MonthlyCharges	High	Customers with fibre-optic service generally pay higher monthly fees. Retaining this pair helps differentiate high-value segments (tech-savvy, higher-income households).
Contract_TwoYear & Tenure	0.56	Longer contracts naturally tie to longer customer tenure. Helps identify loyal, low-churn segments.
Partner & Dependents	0.45	Households with a partner are more likely to have dependents. Captures household structure influencing product usage and churn.
MonthlyCharges & TotalCharges	0.65	TotalCharges accumulates monthly payments, but may include irregularities (discounts, proration). Retaining both captures short-term spending and cumulative value.

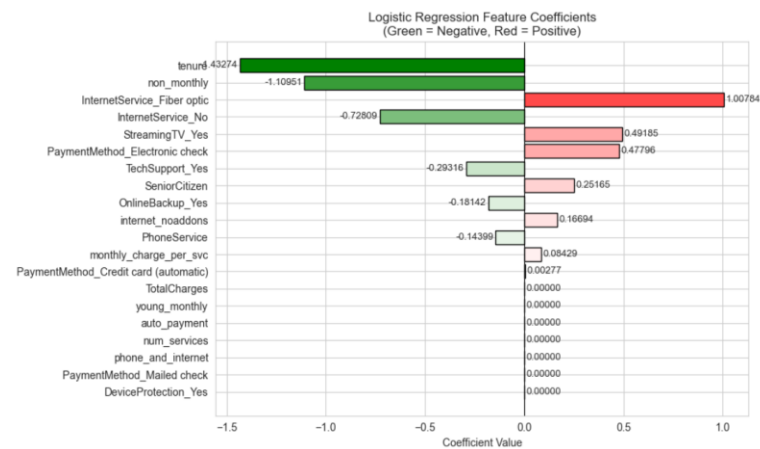
Appendix 3: Churn Prediction Models and Selection Rationale

Model	Rationale	Strengths	Limitations
Logistic Regression	Baseline model to identify and quantify key churn drivers via coefficients.	Simple, interpretable; easy for business stakeholders to understand.	Limited in capturing nonlinear interactions between predictors.
Decision Tree	Segments customers into risk groups using decision rules.	Intuitive, visual; captures simple nonlinear splits (e.g., contract $\times$ tenure).	Prone to overfitting; sensitive to small changes in data.
Random Forest	Reduces overfitting from individual trees; robust to noisy churn data.	Handles multiple predictors; captures nonlinear interactions; less sensitive to imbalance with class weights.	Less interpretable; harder to communicate insights to business users.
Gradient Boosting	Sequentially corrects errors in churn prediction.	Strong predictive accuracy; captures subtle patterns in churn behavior.	Computationally intensive; risk of overfitting if not tuned carefully.
XGBoost	High-performing nonlinear model to identify complex churn patterns.	Handles imbalanced classes and multicollinearity; highlights feature importance.	Less interpretable; requires more hyperparameter tuning.
Light GBM	Efficient gradient boosting for larger feature sets in churn data.	Fast training; can capture nonlinear effects and interactions.	Can overfit small or noisy churn datasets; limited interpretability.
AdaBoost	Focuses on misclassified churn cases to improve recall.	Improves prediction of hard-to-classify churners; interpretable feature importance.	Sensitive to outliers; moderate predictive power for complex churn patterns.
Support Vector Machine (SVM)	Captures complex churn decision boundaries in high-dimensional feature space.	Effective with nonlinear kernels; robust to moderate imbalance.	Computationally expensive; less intuitive for stakeholders.
K-Nearest Neighbour (KNN)	Benchmarks local patterns in churn behavior.	Simple, non-parametric; captures local churn patterns.	Inefficient for large datasets; sensitive to scaling and noise.
Neural Network/ Multi-Layer Perceptron (MLP)	Tests nonlinear and interaction effects for churn prediction.	Flexible; can capture complex dependencies among churn predictors.	Black-box; overfitting risk; harder to translate to business action.

Appendix 4: Complete Model Fitting Results

	Model	Type	Feature_Set	Train_Recall	Test_Recall	Train_AUC	Test_AUC	18	LightGBM	Baseline	Selected	0.83	0.77	0.86	0.83
0	Logistic Regression	Baseline	Selected	0.81	0.82	0.84	0.84	19	LightGBM	Tuned	Selected	0.83	0.80	0.84	0.83
1	Logistic Regression	Tuned	Selected	0.81	0.83	0.84	0.84	20	LightGBM	Baseline	Original	0.83	0.77	0.86	0.83
2	Random Forest	Baseline	Selected	0.83	0.76	0.88	0.82	21	LightGBM	Tuned	Original	0.83	0.80	0.84	0.83
3	Random Forest	Tuned	Selected	0.84	0.78	0.87	0.83	22	Support Vector Machine	Baseline	Selected	0.81	0.79	0.84	0.82
4	Random Forest	Baseline	Original	0.83	0.76	0.88	0.82	23	Support Vector Machine	Tuned	Selected	0.83	0.82	0.83	0.83
5	Random Forest	Tuned	Original	0.84	0.78	0.87	0.83	24	K-Nearest Neighbors	Baseline	Selected	0.86	0.80	0.83	0.77
6	Gradient Boosting	Baseline	Selected	0.81	0.80	0.85	0.84	25	K-Nearest Neighbors	Tuned	Selected	0.86	0.83	0.83	0.79
7	Gradient Boosting	Tuned	Selected	0.82	0.81	0.84	0.84	26	K-Nearest Neighbors	Baseline	Original	0.86	0.80	0.83	0.77
8	Gradient Boosting	Baseline	Original	0.81	0.80	0.85	0.84	27	K-Nearest Neighbors	Tuned	Original	0.86	0.83	0.83	0.79
9	Gradient Boosting	Tuned	Original	0.82	0.81	0.84	0.84	28	Decision Tree	Baseline	Selected	0.83	0.73	0.81	0.74
10	Ada Boost	Baseline	Selected	0.81	0.81	0.84	0.84	29	Decision Tree	Tuned	Selected	0.75	0.72	0.80	0.78
11	Ada Boost	Tuned	Selected	0.89	0.88	0.79	0.79	30	Decision Tree	Baseline	Original	0.83	0.73	0.81	0.74
12	Ada Boost	Baseline	Original	0.81	0.81	0.84	0.84	31	Decision Tree	Tuned	Original	0.75	0.72	0.80	0.78
13	Ada Boost	Tuned	Original	0.89	0.88	0.79	0.79	32	Naive Bayes	Baseline	Selected	0.78	0.76	0.81	0.81
14	XG Boost	Baseline	Selected	0.83	0.74	0.86	0.83	33	Naive Bayes	Tuned	Selected	0.78	0.76	0.81	0.81
15	XG Boost	Tuned	Selected	0.86	0.84	0.83	0.83	34	Naive Bayes	Baseline	Original	0.78	0.76	0.81	0.81
16	XG Boost	Baseline	Original	0.83	0.74	0.86	0.83	35	Naive Bayes	Tuned	Original	0.78	0.76	0.81	0.81
17	XG Boost	Tuned	Original	0.86	0.84	0.83	0.83	36	MLP	Baseline	Selected	0.81	0.79	0.85	0.84
								37	MLP	Tuned	Selected	0.81	0.79	0.85	0.84

Appendix 5 Coefficients of Tuned Lasso-Regularised LR Model



Appendix 6: Complete SHAP Analysis of XGBoost Model

